MOHSIN KHAN

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Professional Summary

Final-year B.Tech student in Artificial Intelligence & Data Science with hands-on experience in machine learning, software development, data analysis, and AI-powered applications. Strong foundation in Python, C++, SQL, model evaluation, and building practical end-to-end systems.

Education

Madhav Institute of Technology & Science (MITS), Gwalior *Expected May 2027*
B.Tech, Artificial Intelligence & Data Science Final-year student
Jawahar Navodaya Vidyalaya, Madhya Pradesh *2022*
Class XII (PCM)

Technical Skills

Languages	Python, C++, SQL, Bash
AI & Machine Learning	PyTorch, Scikit-learn, Hugging Face, NLP, spaCy, OpenCV, MediaPipe
Generative AI & LLMs	LangChain, LangGraph, LlamaIndex, LiteLLM, Ollama, RAG, FAISS
Data & Analytics	NumPy, Pandas, Matplotlib, Seaborn, A/B testing, hypothesis testing
Databases & Tools	PostgreSQL, MongoDB, SQLite, FastAPI, pytest, Docker, Git, Linux

Selected Projects

BERT-Based Sentiment Classification
Python, PyTorch, Hugging Face Transformers, scikit-learn

- Fine-tuned a BERT-based text classifier for sentiment prediction, achieving **93.52% test accuracy** and a reported **ROC-AUC of 0.9675**.
- Monitored training and validation performance to identify overfitting and support careful checkpoint selection.
- Evaluated class-level errors using a confusion matrix (TN=2310, FP=190, FN=134, TP=2366), looking beyond accuracy alone.

NeuroFroge – Neural Network Framework from Scratch
C++, Tensor Operations, Backpropagation, CPU Optimization

- Built a neural-network framework in C++ with N-dimensional tensors, matrix operations, layers, activation functions, cross-entropy loss, backpropagation, and stochastic gradient descent.
- Added mini-batch training, gradient accumulation, tensor broadcasting, and stride-based views, including zero-copy reshape and transpose operations.
- Trained a 784–128–10 Fashion-MNIST classifier, reaching approximately 79% test accuracy in a reported run.
- Profiled and optimized CPU execution; measured approximately 1.76 seconds for a five-epoch experiment compiled with `-O2`.

AI Productivity Assistant
Python, LangGraph, LangChain, Ollama, Playwright, Google Workspace, Telegram

- Developed a modular assistant using a hybrid local/cloud LLM architecture, with local models and cloud inference fallback.
- Used LangGraph to coordinate workflows, human approval for sensitive actions, conversation memory, and token-aware history summarization.
- Integrated Gmail, Calendar, Tasks, Sheets, browser automation, and Telegram file delivery to connect LLM workflows with external tools.

Achievements

- Advanced to Round 2 in Smart India Hackathon, IIIT Delhi Hackathon, and IIIT Gwalior Hackathon.
- Completed the Aspire Leaders Program (2024), focused on leadership development.
- Earned 5-star HackerRank badges in Python, SQL, and C++.

Certifications

- Oracle – Agentic AI Certified Foundations Associate (2026).
- NPTEL – Business Intelligence & Analytics, IIT Madras (79%, Elite).
- freeCodeCamp – Data Analysis with Python (2025).